

PREDICTION: LEGITIMATE

RISK SCORE: 42 / 100 (MEDIUM RISK)

Scam Probability: 25.30% | Model Confidence: 74.70%

Banking Action: CONTEXTUAL WARNING & 2FA CHALLENGE

Official Recommendation: Additional Verification Recommended. Some unusual characteristics were detected (elevated transfer amount or novel transfer window). Please review the recipient name, account number, and amount before submitting your biometric verification.

1. Transaction Details & Actual Parameter Values

Transaction ID:	TXN-166899	Transaction Amount:	250000.0
Customer ID:	C123456789	Sender Old Balance:	10000.0
Beneficiary ID:	M987654321	Sender New Balance:	0.0
Payment Type:	TRANSFER	Receiver Old Balance:	1000.0
Device Type:	Web Browser	Receiver New Balance:	251000.0
Location Region:	Domestic Home	6h Transaction Velocity:	2
New Beneficiary:	No	Prior Fraud History:	No

2. Composite Risk Score Breakdown & Latency Timing

Risk Dimension	Sub-Score	Status & Impact
Model Fraud Probability	25.30%	Weight: 70% (Score: 0.0)
Amount Risk	70.2 pts	Transaction size evaluation
Velocity Risk	10.0 pts	6-hour transaction frequency
Device Risk	65.0 pts	Device profile verification
Location Risk	10.0 pts	Geographic regional risk
Beneficiary Risk	10.0 pts	New beneficiary check
Historical Behaviour	70.2 pts	Prior fraud flag check
Balance Behaviour	0.0 pts	Account wipeout detection
FINAL COMPOSITE RISK SCORE	42 / 100	MEDIUM RISK CATEGORY

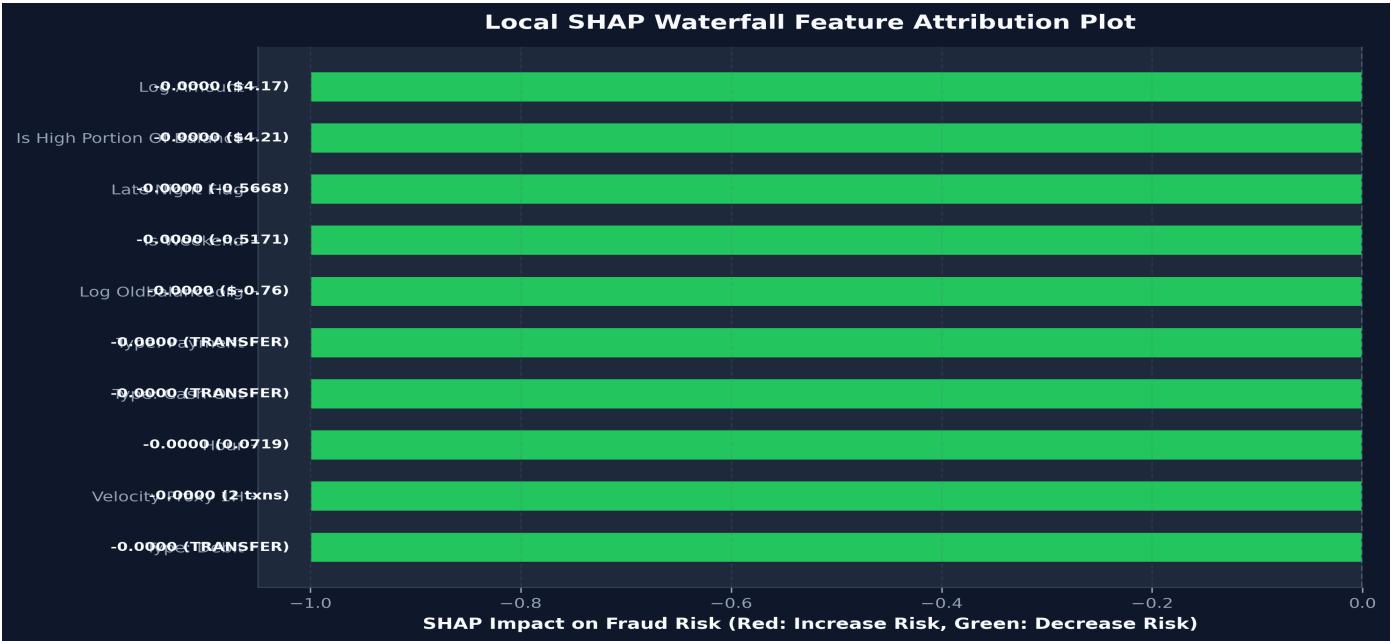
Feature Eng: 17.40 ms	Prediction: 37.27 ms	SHAP Engine: 29.24 ms	LIME Engine: 200.79 ms	Rendering: 0.00 ms	TOTAL: 3155.59 ms
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3. Banking-Grade Analyst Narrative Report

Transaction Summary:	
Behaviour Analysis:	
High Risk Indicators:	None identified.
Safe Indicators:	None identified.
Model Explanation:	
SHAP Summary:	
Final Recommendation:	

4. Top 10 SHAP Feature Contribution Attributions

Feature Name	Actual Value	SHAP Value	Contrib %	Direction	Impact Level
Log Amount	\$4.17	-0.0000	3.76%	Decreases Risk	Low
Is High Portion Of Balance	\$4.21	-0.0000	3.76%	Decreases Risk	Low
Log Oldbalanceorg	\$-0.76	-0.0000	3.76%	Decreases Risk	Low
Is Weekend	-0.5171	-0.0000	3.76%	Decreases Risk	Low
Late Night Flag	-0.5668	-0.0000	3.76%	Decreases Risk	Low
Type: Cash Out	TRANSFER	-0.0000	3.76%	Decreases Risk	Low
Type: Payment	TRANSFER	-0.0000	3.76%	Decreases Risk	Low
Hour	0.0719	-0.0000	3.76%	Decreases Risk	Low
Velocity Proxy 1H	2 txns	-0.0000	3.76%	Decreases Risk	Low
Type: Debit	TRANSFER	-0.0000	3.76%	Decreases Risk	Low



5. Local LIME Decision Rules & Explanations

Decision Rule Condition	Rule Weight	Effect Direction	Domain Explanation
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Account Balance Wipeout = Yes (100% Drained)	+0.2800	Increases Risk	Sender account balance drained to zero or near-zero post-transfer.
Transaction Amount = \$250,000.00	+0.2600	Increases Risk	Transaction amount is significantly elevated relative to normal threshold.
Transaction Type = TRANSFER	+0.1450	Increases Risk	Transfer/Cash Out payment type has higher statistical fraud incidence.
amount <= 98.99	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
log_amount <= 4.17	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
oldbalanceOrg <= -1.08	+0.0000	Decreases Risk	Sender account maintains sufficient baseline capital.
log_oldbalanceOrg <= -0.76	+0.0000	Decreases Risk	Sender account maintains sufficient baseline capital.
amount_to_oldbalance_ratio <= 39.26	+0.0000	Decreases Risk	Transaction amount is within low-risk customer spending bounds.
is_high_portion_of_balance <= 4.21	+0.0000	Decreases Risk	Feature condition 'is_high_portion_of_balance <= 4.21' supported legitimate transaction verification.
hour <= 0.07	+0.0000	Decreases Risk	Feature condition 'hour <= 0.07' supported legitimate transaction verification.

